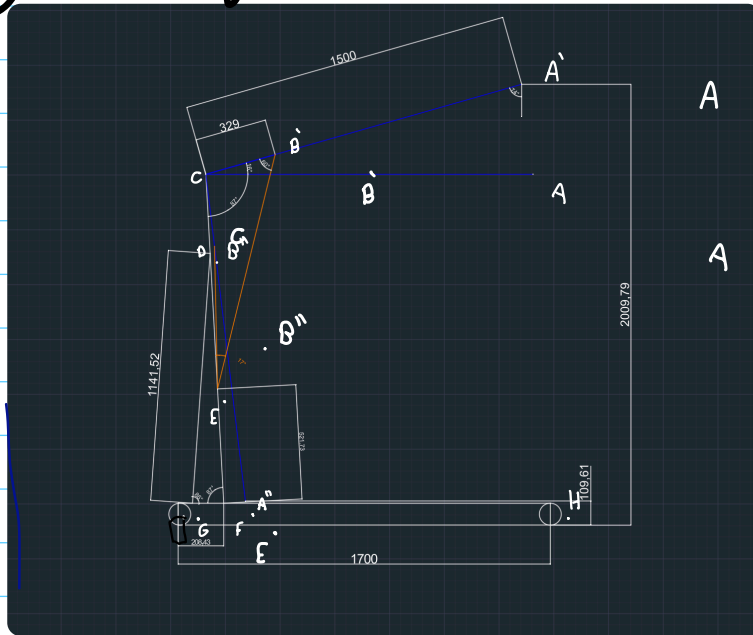


Engineering Design I, Project, Milestone 1



- Equilibrium equations:-

$$\rightarrow \sum F_y = 0; 9810 = B \sin(76) - C_y$$

$$\rightarrow \sum F_x = 0; C_x = B \cos(76)$$

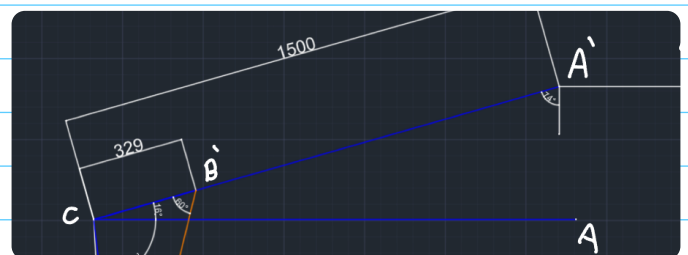
$$\rightarrow \sum M_C = 0; 9810 \sin(74) \times 1500 + B \cos(76) \times 329 \times \sin(16)$$

=

$$\rightarrow B \sin(76) \times 329 \times \cos(16)$$

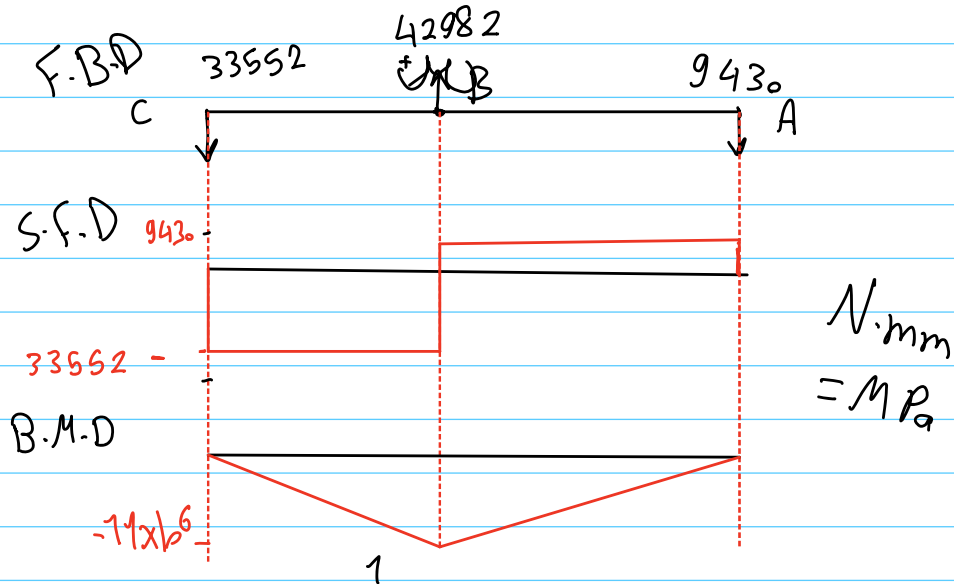
$$\Rightarrow B(285) = 14144966$$

$$\hookrightarrow \underline{\underline{B = 49631.5}}$$



- $C_x = 49631.5 \cos(76) = 12007$

- $C_y = 49631 \sin(76) - 9810 = 38347$

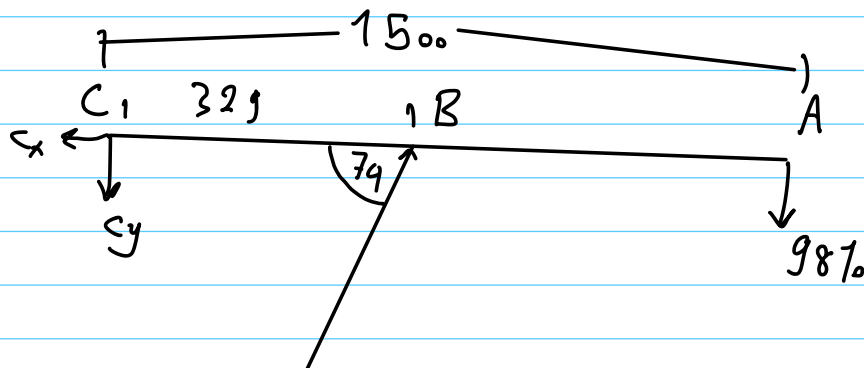


$$\rightarrow \sum F_y = 0; 9810 + C_y = B \sin(74)$$

$$\rightarrow \sum F_x = 0; C_x = B \cos(74)$$

$$\rightarrow \sum M_C = 0; 9810 \times 1500 = B \sin 74 \times 329$$

$$\Rightarrow \underline{\underline{B = 46529}}$$



$$CD = 361.1$$

$$C_x = 12825$$



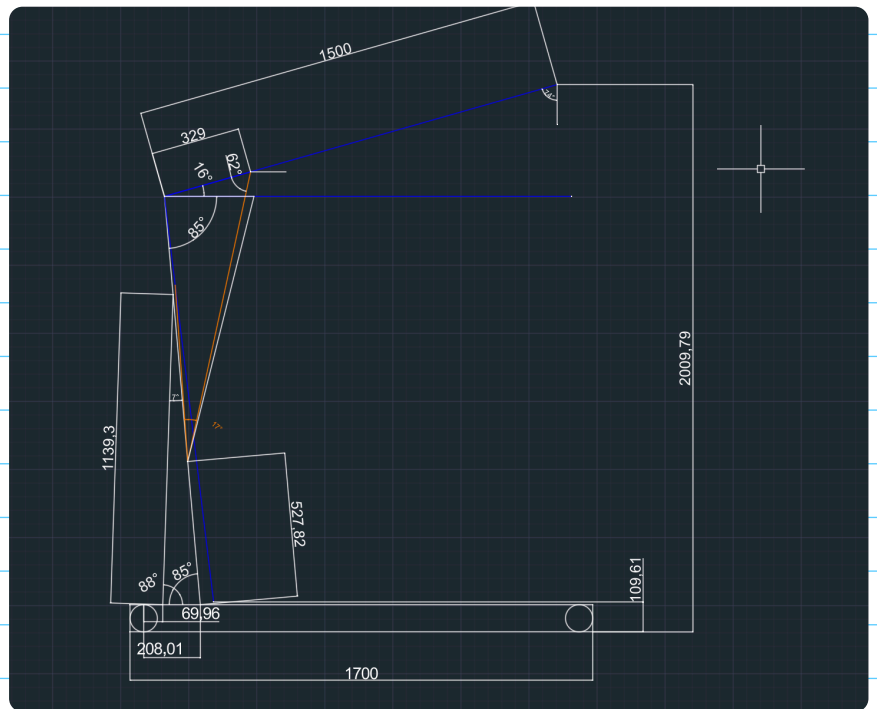
$$C_y = 34917$$



$$2D \cos 88 = F_x$$

$$\rightarrow 0.0698D$$

$$\rightarrow F = 0.8D$$

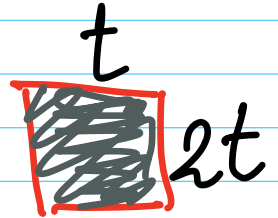


Stress Analysis:-

Ac: B is the critical point of application

$$\rightarrow \sigma_b = \frac{Mc}{I} = \frac{-11 \times 10^6 \times t}{\frac{1}{12}(b h_o^3 - b_i h_i^3)} = \frac{39.4 \times 10^6}{t^3}$$

$$\rightarrow \sigma = \frac{F}{A} + \frac{Mc}{I} = \frac{8.28}{0.56 t^2} - \frac{39.4 \times 10^6}{t^3}$$

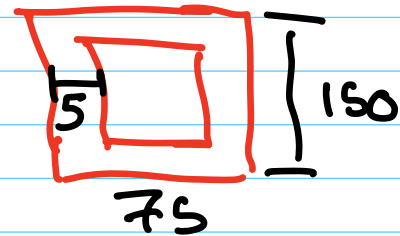


$$\rightarrow 2 = \frac{\sigma_y}{\sigma}$$

$$\rightarrow 137.5 = \frac{8.22}{0.56 t^2} - \frac{39.9 \times 10^6}{t^3}$$

$$\rightarrow \sigma = 137.5, \quad t = 66$$

$$\rightarrow \sigma'_b = \frac{11 \times 10^6 \times 75}{6.23 \times 10^6}$$



$$\rightarrow \frac{F}{A} = \frac{-9810 (\cos(74))}{(150)(75) - (140)(65)} = 1.26 \text{ MPa}$$

$$\rightarrow I' = \frac{1}{12} (75)(150)^3 - \frac{1}{12} (65)(140)^3$$

$$\rightarrow I' = 6.23 \times 10^6 \text{ mm}^4$$

$$\rightarrow \sigma' = \frac{F}{A} + \sigma'_b = 131.166 \text{ MPa}$$

→ Type of steel:-

→ (150 x 75 x 5) mm

→ S27 Steel hollow steel section
(16.37)

$$\rightarrow B(285) = 14144966$$

$$\rightarrow B = 49631.5 \text{ N}$$

$$\rightarrow C_x = 49631.5 \cos(76) = 12007$$

$$\rightarrow C_y = 49631 \sin(76) - 9810 = 38347$$

$$\rightarrow \sum F_y = 0; F_y + D \sin(86) = C_y + E \sin(78) \quad (1)$$

$$\rightarrow -F_y + D \sin(86.15) = 86894$$

$$\rightarrow 2 \cos(86) D = F_x$$
$$\rightarrow 0.196 D = F_x$$

$$\rightarrow \sum F_x = 0; C_x + E \cos(78) = D \cos(86.15) + F_x$$

$$\rightarrow 22326 = D \cos(86.15) + F_x$$

$$\rightarrow 22326 = 0.263 D$$

$$\rightarrow D = 88016.7$$

$$\rightarrow F_x = 14874.8$$

$$\rightarrow \sum M_F = M_F$$

$$\rightarrow E \sin(77) \times 527.82 + 15213 \times 1505 - D \sin(9.87) \times 802.7 = M_F$$

$$\rightarrow M_F = 18594756 \text{ N}\cdot\text{mm}$$

$$C F_y: 807.6 (\downarrow)$$

$$F_x = 14874.8 (\rightarrow)$$

$$= 88016.7$$

$$CE = 977$$

$$CD = 703.4$$

$$CF = 1505$$

$$\rightarrow C = \sqrt{C_y^2 + C_x^2} = \sqrt{38347^2 + 12007^2}$$

$$F = 14897$$

$$\theta = 3.1$$

$$B = 46529$$

$$C = 40183$$

$$D = 88016.7$$

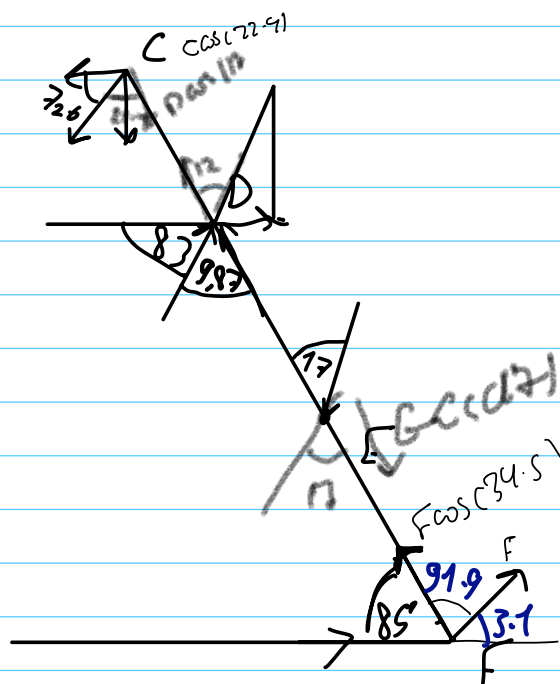
$$\rightarrow F = \sqrt{F_y^2 + F_x^2} = 14897 = 40183$$

$$\rightarrow \tan^{-1}\left(\frac{38347}{12007}\right) = 72.6$$

$$\rightarrow \alpha = 60.52$$

$$\rightarrow F \sin(34.5) + 30249.5 \sin(34.5) = 17133.5$$

$$\rightarrow \underline{C_{\perp}} = 40183 \sin(22) = 15313$$



$$G + H = D \sin \alpha + F_y$$

$$G + H = 8000$$

$$F_y = D \sin \alpha - 86894$$

$$D = \frac{22326}{3 \cos \alpha} =$$

$$F_y = 7442 \tan \alpha - 86894 = -39960$$

$$G + H = 14884 \tan \alpha - 86894 = 8000$$

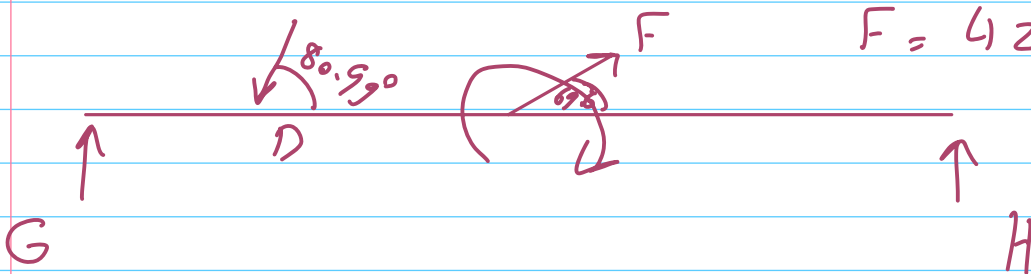
$$14884 \tan \alpha = 94894$$

$$\alpha = 80.99$$

$$\begin{aligned} F_x &= D 2 \cos(\alpha) \\ &= 47520.3 \times 2 \times \cos(80.99) \\ &= 14884 \end{aligned}$$

$$G + H = 6974$$

$$F = 42642 \quad \theta = 69.6$$



$$B = 49631.5 \text{ N}$$

$$\rightarrow \Sigma M_F = M_F$$

$$\rightarrow E \sin(17) \times 527.82 + 15313 \times 1505.08 \sin(9.87) \times 802.7 = M_F$$

$$\rightarrow M_F = 18594756 \text{ N}\cdot\text{mm}$$

$$49631.5 \sin(17) \times 527.82 + 15313 \times 1505$$

$$= 47520.3 \times \sin(14.04) \times 561.5$$

$$= 24231989.9$$

$$47520.3$$

$$\Sigma M_F = 0;$$

$$-G \times 208 + 47520.3 \times \sin(80.99) \times 138.04$$

$$-24231989.9 + (6974 - G) \times 1392$$

$$= 0$$

$$1600 G = -8045419.7$$

$$G = 5028$$

$$H = 6974 - G = 1946$$

